

1. Let f, g be two functions defined from $D \subset \mathbb{R}$ into $\mathbb{R}$. Translate using quantifiers the following statements (i.e : Write in mathematical language) :

- 1) f is bounded ; 2) f is odd ; 3) f is never equal to 0 ; 4) f is periodic ;
5) f is increasing ; 6) f is not the 0 function ; 7) f does not have the same value at two different points ; 8) f is not less than g .

2. Determine the domain of definition for the following functions.

$$f_1(x) = \sqrt[4]{x - x^3}, \quad f_2(x) = \ln(x - 3) - \ln(x), \quad f_3(x) = \ln\left(1 - \frac{3}{x}\right),$$
$$f_4(x) = \ln(1 + \lfloor x \rfloor), \quad f_5(x) = \ln(\ln(x)).$$

3. Show, by terms of definition, that the following functions are increasing (decreasing) on the specified set E .

$$f_1(x) = \ln\left(1 + \frac{1}{x}\right), \quad E =]0, +\infty[.$$
$$f_2(x) = \cos(x), \quad E = [0, \pi].$$
$$f_3(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad E = \mathbb{R}.$$
$$f_4(x) = \lfloor x \rfloor (x + \lfloor x \rfloor)^{\lfloor x \rfloor}, \quad E = [1, +\infty[.$$

4. Study the parity of the following functions :

$$f_1(x) = 3x - x^3 + 3x^7, \quad f_2(x) = \sqrt[5]{(1-x)^2} + \sqrt[5]{(1+x)^2}, \quad f_3(x) = a^{\sin(x)} + a^{-\sin(x)} \quad (a > 0),$$
$$f_4(x) = \ln\left(\frac{4-x}{4+x}\right), \quad f_5(x) = \ln\left(x + \sqrt{1+x^2}\right).$$

5. By using (ε, δ) definition of the limit, show that

$$\lim_{x \rightarrow -2} x^2 - 1 = 3, \quad \lim_{x \rightarrow -2} \frac{x^2 + x + 1}{1 - x} = 1, \quad \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty, \quad \lim_{x \rightarrow a} \sqrt{x} = \sqrt{a}, \quad a \geq 0.$$

6. Find the following limits or say if they do not exist.

(a) $\lim_{x \rightarrow 0} x \cos \frac{1}{x}$, (b) $\lim_{x \rightarrow 0} x \left\lfloor \frac{1}{x} \right\rfloor$, (c) $\lim_{x \rightarrow 0} \frac{x}{a} \left\lfloor \frac{b}{x} \right\rfloor$, $a, b > 0$,
(d) $\lim_{x \rightarrow 0} \frac{\lfloor x \rfloor}{x}$, (e) $\lim_{x \rightarrow +\infty} x (\sqrt{x^2 + 1} - \sqrt[3]{x^3 + 1})$, (f) $\lim_{x \rightarrow 0} \frac{\cos(\frac{\pi}{2} \cos x)}{\sin(\sin x)}$.

7. Show that the function $f(x) = \cos(\ln(x))$ does not have limits at $+\infty$.

8. Find the following limits.

$$\lim_{x \rightarrow +\infty} \left(x - \sqrt{x^2 - 2x} \right), \quad \lim_{x \rightarrow +\infty} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{x + \sqrt{x}}}, \quad \lim_{x \rightarrow 0} x \sin \left(\frac{1}{x} \right),$$

$$\lim_{x \rightarrow 0^+} \left(1 + \sin^2 \frac{1}{x} \right) \ln x, \quad \lim_{x \rightarrow 0} x \left\lfloor \frac{1}{x} \right\rfloor, \quad \lim_{x \rightarrow 0^\pm} \frac{x - \lfloor x \rfloor}{\sqrt{|x|}}, \quad \lim_{x \rightarrow 0} \frac{\lfloor 1/x \rfloor + x}{\lfloor 1/x \rfloor - x}.$$

9. For $a > 1$ and $\alpha > 0$, show that

$$\lim_{x \rightarrow +\infty} \frac{a^x}{x^\alpha} = +\infty, \quad \lim_{x \rightarrow +\infty} \frac{\ln x}{x^\alpha} = 0, \quad \lim_{x \rightarrow 0^+} x^\alpha \ln x = 0.$$

10. Prove that if $f \in C[a, b]$ then $|f| \in C[a, b]$. Show by means of example that the converse is not true in general.

11. Study the continuity on $\mathbb{R}$ of the following functions.

$$f(x) = \lfloor x \rfloor + \sqrt{x - \lfloor x \rfloor}, \quad g(x) = x \mapsto \lfloor x \rfloor - (x - \lfloor x \rfloor)^2.$$

12. Find all the points where the function f defined by

$$f(x) = \begin{cases} x^2 - 1 & \text{if } x \text{ is irrational} \\ 0 & \text{if } x \text{ is rational} \end{cases}$$

is continuous

13. Study the uniform continuity of the following functions :

$$f_1(x) = \frac{x}{4 - x^2} \quad (-1 \leq x \leq 1), \quad f_2(x) = \ln x \quad (0 < x < 1), \quad f_3(x) = \frac{\sin x}{x} \quad (0 < x < \pi),$$

$$f_4(x) = 2 \sin x - \cos x \quad (x \in \mathbb{R}), \quad f_5(x) = \frac{1}{x} \quad (0.1 < x < 1).$$

14. Show that any continuous periodic function $f : \mathbb{R} \rightarrow \mathbb{R}$ is uniformly continuous.

15. (a) Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous, periodic and non-constant function, then it admits a smaller strictly positive period called the primitive period.

(b) Give an example of a periodic and non-constant function not admitting a primitive period.

(c) Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a periodic function without a primitive period, then the set of its periods is dense in $\mathbb{R}$.

16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = \sin(x^2)$. Prove that :

(a) f is uniformly continuous on any interval $[0, a]$ with $a > 0$,

(b) f isn't uniformly continuous on $[0, +\infty)$.

17. Let $F : (0, +\infty) \rightarrow (0, +\infty)$ be a continuous function satisfying

$$\lim_{x \rightarrow +\infty} \frac{F(x)}{x} = k > 1 > l = \lim_{x \rightarrow 0^+} \frac{F(x)}{x}.$$

Prove that the equation $F(x) = x$ has a solution.